

Day 5: Case Study Practical

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

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1. Check the power outlet

Plug another device into the same outlet to confirm that the outlet is providing power.

2. Check the charger and cable

Inspect the charger, cable, and connector for cuts, burns, loose connections, or other damage. Check whether the charging indicator turns on.

3. Check the charging port

Inspect the laptop's charging port for dust, damage, or looseness. Make sure the charger fits firmly into the port.

4. Perform a power reset

Disconnect the charger and turn the laptop off. If the battery is removable, remove it. Hold the Power button for 15–20 seconds, reconnect the charger, and try starting the laptop again.

5. Check the battery/charger with a known-good compatible charger

If available, try another compatible charger. If the laptop powers on with it, the original charger may be faulty. If it still does not power on, the battery, charging circuit, or motherboard may require professional inspection.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue. Hint: Consider checking both hardware connections and basic display settings.

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Step-by-Step Troubleshooting Flow

Step 1: Monitor Power & Input Source Verification: Checking power, LED states, and input sources.

- 1. Check Monitor Power:** Verify that the monitor's power LED is illuminated (solid blue/white indicates active power; amber/orange indicates stand-by mode).
- 2. Cycle Power:** Turn the monitor OFF, wait 5 seconds, and turn it back ON. Watch for splash screens (e.g., Dell, LG, ASUS logo). If no logo appears, the monitor itself may be faulty or unplugged from power.
- 3. Check Input Source:** Press the Input/Source button on the monitor controls to manually cycle through inputs (HDMI 1, HDMI 2, DisplayPort). Monitors often fail to switch automatically if they enter deep power-save mode.

Step 2: Physical Cable & GPU Port Inspection: Verifying physical display cables and port selection.

1. **Check Cable Connections:** Ensure both ends of the HDMI/DisplayPort cable are firmly seated into the monitor and the PC tower.
2. **Verify Dedicated vs. Integrated Port (Crucial):**
 - If the PC has a dedicated graphics card (GPU), ensure the display cable is plugged directly into the lower horizontal GPU ports.
3. **Test Cable Integrity:** Swap the cable with a known-working HDMI or DisplayPort cable, or test with a different port type (e.g., switch from DisplayPort to HDMI).

Step 3: Operating System & Display Driver Reset: Determining if the Operating System is booting behind the scenes.

1. **Audio/Keyboard Check:** Press the Caps Lock or Num Lock keys on the keyboard to check if the indicator LEDs toggle ON/OFF. If they toggle, the motherboard and OS are running fine in the background.
2. **Reset Display Driver:** Press the Windows shortcut keys Win + Ctrl + Shift + B. The screen will flash, a short beep will sound, and Windows will attempt to refresh the display driver.
3. **Multi-Monitor Settings:** Press Win + P to open the display projection menu, then press the Down Arrow and Enter to cycle through display modes (Duplicate/Extend).

Step 4: Internal Hardware Inspection (Reseating Components): Checking system memory and internal expansion cards.

1. Shut down the PC completely, switch off the Power Supply Unit (PSU), and unplug the power cable.
2. Open the side panel of the computer chassis.

Step 5: Hardware Isolation & Final Resolution: Isolating hardware components to confirm the fault.

1. **Test Monitor Elsewhere:** Connect the monitor to another laptop or PC. If it still displays a blank screen, the monitor itself is defective and requires replacement.
2. **Clear CMOS:** Remove the motherboard coin-cell battery for 5 minutes to reset BIOS settings back to factory defaults (resolving bad display/RAM frequency configurations).
3. **Escalation:** If all physical tests pass and the monitor works on another device, inspect the PC for a faulty GPU, failing Power Supply Unit, or defective motherboard.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

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1. Check the mouse power

Make sure the mouse is switched **ON**. Replace the batteries or charge the mouse if the battery is low. Check whether the mouse's indicator light turns on.



2. Check the wireless receiver

If the mouse uses a USB receiver, remove it and plug it back in firmly. Check whether the receiver is damaged.



3. Test another USB port

Move the mouse receiver to a **different USB port**.

- If it works → the original USB port may have a problem.
- If it still doesn't work → continue troubleshooting.



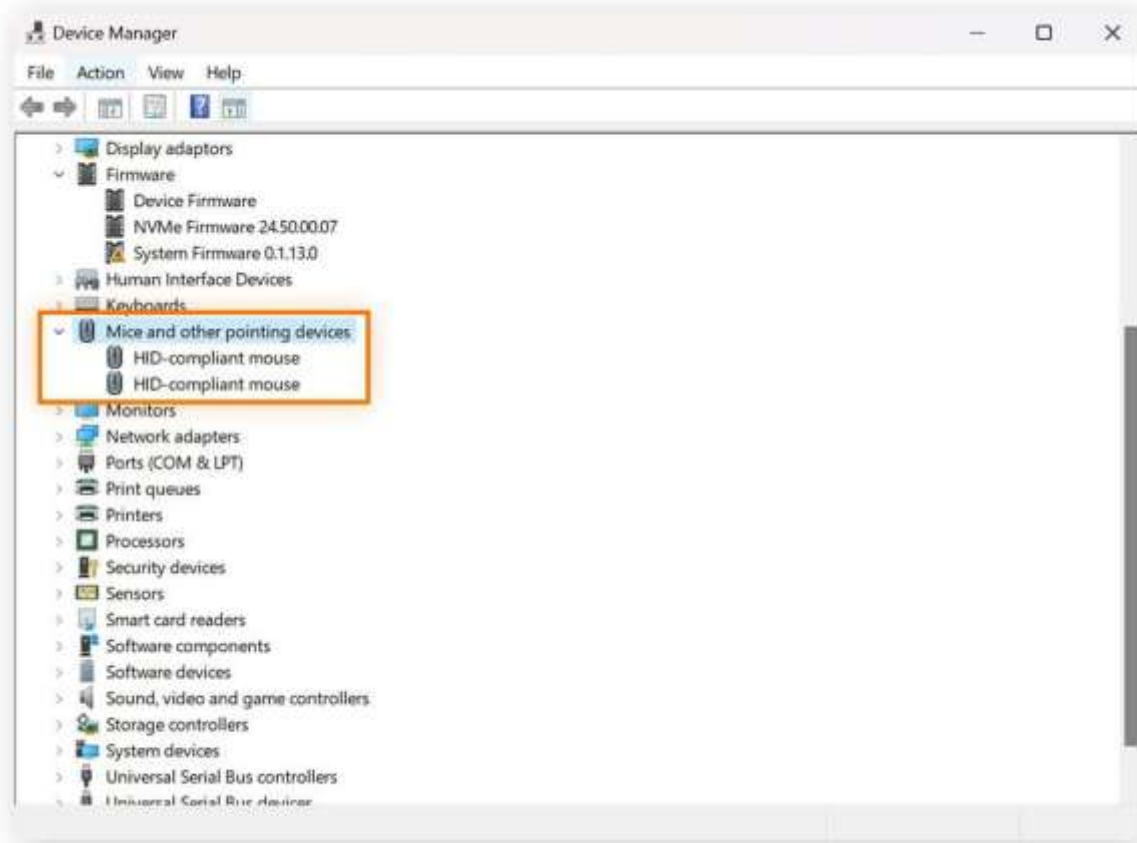
4. Test the mouse on another computer

Connect the wireless receiver to another computer and test the mouse.

- **Works on another PC** → the original computer's USB port or system settings may be the problem.
- **Doesn't work on another PC** → the mouse, receiver, or batteries are likely faulty.

5. Check Windows settings and Device Manager

Use the keyboard to open **Device Manager** and check **Mice and other pointing devices** or **Human Interface Devices**. Look for a warning symbol or disabled device. Enable or reinstall the device if necessary.



6. Check Bluetooth settings (if it is a Bluetooth mouse)

Make sure **Bluetooth is enabled** and the mouse is paired and connected. Remove the mouse from Bluetooth devices and pair it again if necessary.

7. Restart the computer

Restart Windows and test the mouse again. This can resolve temporary USB or driver problems.

Final Diagnosis

- ❖ **Mouse fails on every computer** → likely mouse/receiver or battery problem.
- ❖ **Mouse works in another USB port** → original USB port may be faulty.
- ❖ **Mouse works on another computer but not this PC** → likely a driver or system-settings problem.
- ❖ **Mouse works after reconnecting/pairing** → connection problem was the cause.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one. Hint: Think about fans, dust, and thermal paste.

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1. Faulty or blocked cooling fans

Possible cause: The CPU or GPU fan may not be spinning properly, or a fan could be blocked.

How I would check:

- Shut down the PC and disconnect the power.
- Open the case and visually inspect the **CPU, GPU, and case fans**.
- Check for fans that are not spinning freely or making unusual noises.
- Make sure cables are not blocking the fan blades.
- Replace a faulty fan if necessary



2. Dust buildup

Possible cause: Dust can block the heatsink and air vents, reducing airflow and causing the components to overheat.

How I would check:

- Turn off and unplug the PC.
- Inspect the **CPU heatsink, GPU heatsink, fans, and case air vents**.
- Look for thick layers of dust blocking airflow.
- Clean the dust carefully using **compressed air**.

- Make sure the case has good airflow after cleaning.



3. Old or improperly applied thermal paste

Possible cause: Thermal paste between the CPU and heatsink can dry out or become ineffective, reducing heat transfer from the CPU to the heatsink.

How I would check:

- Power off and unplug the PC.
- Remove the CPU cooler carefully.
- Inspect the thermal paste between the CPU and cooler.
- If it is **dry, cracked, or uneven**, clean the old paste with appropriate isopropyl alcohol and apply a small amount of new thermal paste.
- Reinstall the cooler evenly and securely.



